COMMUNITY HERO: HYPERLOCAL CIVIC PROBLEM SOLVER & SMART CITY INCIDENT RESOLUTION PLATFORM WITH 3D GEOSPATIAL VISUALIZATION AND MACHINE LEARNING

PROJECT REPORT [Phase-III]

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

Department of Computer Science and Engineering

Submitted to

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL (M.P.)



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October 2026

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
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CERTIFICATE

This is to certify that the work embodied in this report entitled "Community Hero: Hyperlocal Civic Problem Solver & Smart City Incident Resolution Platform with 3D Geospatial Visualization and Machine Learning" has been satisfactorily completed by AARUSH BHARTI (24U020010) & ASHISH GOUTAM (24U020011). It is a bonafide piece of work, carried out under our guidance in the Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhopal for the partial fulfillment of the Bachelor of Technology during the academic year 2026-27.

The web application, trained classification model, 75-meter spatial duplicate detection algorithm, autonomous multilingual voice dispatch pipeline, and testing phases have been systematically evaluated and validated as part of the Phase III minor project deliverables.

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DECLARATION

We hereby declare that the following project report entitled "Community Hero: Hyperlocal Civic Problem Solver & Smart City Incident Resolution Platform with 3D Geospatial Visualization and Machine Learning" presented in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is an authentic documentation of our original work, carried out under the able guidance of Dr. Sreemoyee Biswas. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole for the award of any degree or professional diploma in any other institute or organization.

The software codebase, 3D digital twin rendering, trained machine learning model, autonomous voice dispatch engine, database architecture, and empirical experimental results presented in this report were developed and evaluated by us. All external reference materials, standard formulas, and open-source packages have been formally cited in accordance with IEEE formatting standards.

We declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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ACKNOWLEDGEMENT

We, Aarush Bharti and Ashish Goutam, would like to express our sincere gratitude to everyone who supported and guided us throughout the successful completion of our project titled “Community Hero: Hyperlocal Civic Problem Solver & Smart City Incident Resolution Platform with 3D Geospatial Visualization and Machine Learning”.

First and foremost, we extend our heartfelt thanks to our esteemed Director, Dr. Ashutosh Kumar Singh, for providing the state-of-the-art laboratory resources and academic framework needed to carry out this project.

We are deeply grateful to our Project Guide, Dr. Sreemoyee Biswas, Assistant Professor, Department of Computer Science and Engineering, for her constant guidance, valuable suggestions, and expert advice. Her insights helped us refine our technical concepts, algorithmic formulations, and improve our system design, while her support kept us motivated throughout challenging phases.

Our sincere thanks to the Department Coordinator, Dr. Rahul Chaurasia, for the support and encouragement provided during the completion of the project. We also thank all the faculty members of the CSE department for clarifying our doubts and providing continuous direction.

A special thanks to our batchmates and friends for their continuous support, code review suggestions, and active participation in user-testing the 3D web interface and multilingual voice dispatch system. Last but not least, we express our deepest gratitude to our parents and family members for their unconditional love, patience, and moral support throughout our undergraduate coursework.

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1. ABSTRACT

In modern urban governance, managing everyday civic infrastructure grievances—such as road potholes, shattered streetlights, rupturing water mains, unattended solid waste piles, and hazardous downed electrical wires—remains deeply compromised by antiquated manual reporting channels. Municipal portals and telephonic helplines often deliver opaque, one-way interactions that alienate citizens, while civic administrations are overwhelmed by hundreds of unstructured, duplicate reports that trigger uncoordinated, costly field deployments. To solve this systemic civic friction, this project presents Community Hero, an autonomous, full-stack civic problem solver and smart city incident resolution platform.

Community Hero bridges the physical-digital divide using an interactive 3D WebGL Digital Twin (Three.js) coupled with high-resolution 2D Leaflet street mapping, enabling residents to pinpoint geospatial coordinates, upload real-time photographic proof, and monitor resolution telemetry with sub-second feedback. To eliminate reliance on expensive commercial cloud APIs, the system incorporates an offline Natural Language Processing (NLP) pipeline combining TF-IDF feature extraction with a Multinomial Naive Bayes classifier trained on 1,200 curated municipal grievance records across five core categories, reaching 89.2% classification accuracy with an average inference latency under 5 milliseconds. The backend pairs classification with a deterministic hazard severity matrix that assigns urgency scores from 1 to 5 based on acute municipal safety risks.

To combat catastrophic 911/311 phone-bank surges during civic emergencies, a 75-meter spatial deduplication engine powered by the Haversine formula clusters proximate reports into single unified incident parents, consolidating crowd-sourced confirmations and incrementing priority upvotes. Furthermore, an autonomous voice AI module powered by Bolna and Web Speech APIs provides real-time telephonic and browser microphone intake in Hindi and English with live waveform visualization, transcript parsing, and text-to-speech audio feedback. The backend implements the international Open311 v2 GeoReport standard with Server-Sent Events (SSE) for zero-latency dashboard alerts. Rigorous empirical evaluations demonstrate sub-25ms response latency under concurrent municipal load, verifying that Community Hero offers a scalable, equitable foundation for next-generation smart city administration.

Keywords: Smart Cities, Civic-Tech, 3D Digital Twin, Three.js, React, Node.js, SQLite, Machine Learning, Offline NLP, Haversine Deduplication, Bolna Voice AI, Open311 GeoReport v2.

2. INTRODUCTION

2.1 Background and Real-World Context

Rapid urbanization across the globe has placed unprecedented strain on municipal governance systems. In metropolitan areas and tier-2 smart cities, basic civic infrastructure breakdowns—such as dangerous road potholes, malfunctioning street lighting, leaking water mains, open drainage trenches, garbage overflow, and sparking electrical transformers—represent daily safety hazards and quality-of-life bottlenecks. In developing economies such as India, these hazards directly impact public health and economic productivity, causing vehicular damage, traffic gridlock, and preventable fatal accidents.

Historically, municipal municipal corporations rely on centralized grievance redressal mechanisms, including telephone helplines (such as 311 or CM Helplines), municipal ward office desks, and static online web portals. However, these traditional channels exhibit profound structural weaknesses: they are largely paper-based or text-heavy, offer no real-time transparency into ticket resolution stages, and force citizens to navigate confusing bureaucratic hierarchies without geographical context.

2.2 Hyperlocal Civic Issue Reporting Bottlenecks

From the municipal administrative perspective, the inbound reporting bottleneck is equally catastrophic. During seasonal heavy rains, monsoons, or extreme weather events, dozens or even hundreds of independent citizens may report the exact same collapsed roadway or fallen electrical pole within minutes. Traditional civic software lacks automated spatial correlation, treating each incoming phone call or web form as an isolated incident. Consequently, dispatch officers inadvertently assign multiple municipal repair trucks, field engineers, and contractors to the same geographical site, rapidly draining limited municipal budgets and leaving other critical zones unattended.

2.3 Why Voice AI and Multilingual Accessibility

A major deficiency of modern digital civic portals is the systemic exclusion of non-tech-savvy, elderly, or illiterate citizens who cannot navigate complex smartphone user interfaces or type descriptive complaints in English. In multilingual nations, linguistic barriers severely disenfranchise vulnerable demographics from civic participation. Implementing conversational Voice AI capable of understanding colloquial spoken Hindi, English, and Hinglish via standard telephonic calls or one-click browser microphone interaction democratizes civic reporting. Callers can simply narrate their problem naturally, while autonomous speech processing models extract the incident nature, landmark location, and

severity level, automatically converting unstructured speech into actionable, structured municipal work orders.

2.4 Project Motivation

Core Objectives & Philosophy: As undergraduate computer science researchers at the Indian Institute of Information Technology, Bhopal, we were motivated to design and implement an end-to-end, privacy-preserving, and computationally lightweight platform that resolves these civic communication failures. The core motivations behind Community Hero are:

- **1. Spatial Immersion:** Engaging 3D Visual Experience: Replace archaic tabular portals with an immersive 3D WebGL Digital Twin Earth combined with 2D Leaflet street maps, providing citizens and city administrators with an immediate spatial mental model of municipal health.
- **2. Cost & Privacy:** Offline Edge Machine Learning: Eliminate recurring cloud API costs by building and training a local, privacy-preserving NLP classifier that sorts civic complaints and calculates urgency within 5 milliseconds.
- **3. Resource Optimization:** Swarm-Surge Deduplication: Formulate and deploy a 75-meter Haversine spatial duplicate clustering algorithm that groups redundant citizen reports into unified incident trees, stopping wasted municipal dispatch expenditures.
- **4. Universal Accessibility:** Conversational Voice AI Accessibility: Bridge the digital divide through telephony and browser-based voice agents supporting English and Hindi speech intake with closed-loop CAD dispatch.

2.5 Scope and Key Contributions

Delivered Scope: The deliverables successfully achieved in Phase III of this project encompass:

- • **Full-Stack Architecture:** A full-stack reactive web portal built with React 18, Vite, Three.js 3D Globe, and Leaflet 2D street map.
- • **High-Performance Backend:** An Express.js backend server backed by SQLite in Write-Ahead Logging (WAL) mode for resilient, zero-configuration local database operations.
- • **Machine Learning Engine:** A trained Python Scikit-Learn NLP text classification pipeline achieving 89.2% accuracy across 5 civic categories.
- • **Spatial Deduplication:** A real-time 75-meter spatial duplicate detection algorithm executing in sub-5ms over SQLite geometry records.

- • **Multilingual Voice CAD:** An autonomous voice intake engine supporting telephonic Bolna agents and client-side Web Speech recognition with speech synthesis feedback.
- • **Standards & Telemetry:** Full compliance with the international Open311 GeoReport v2 REST specification and Server-Sent Events (SSE) telemetry.

2.6 Organisation of the Report

The remainder of this report is structured as follows: Chapter 3 reviews related literature and smart city frameworks. Chapter 4 provides the formal problem definition, mathematical models, and success criteria. Chapter 5 describes the methodology, dataset, machine learning pipeline, and system architecture. Chapter 6 details the core algorithms and computational complexity. Chapter 7 presents system diagrams, DFDs, and visual interfaces. Chapter 8 outlines implementation details, software stack, and code listings. Chapter 9 presents comprehensive empirical results, testing scenarios, and latency benchmarks. Chapter 10 concludes the report and discusses future research horizons. Chapters 11, 12, and 13 document references, publications, and plagiarism verification.

3. LITERATURE REVIEW

3.1 Existing Municipal 311 Grievance Systems

Crowdsourced civic problem reporting gained prominence with platforms such as FixMyStreet in the United Kingdom and SeeClickFix in North America [1], [3]. These platforms demonstrated that empowering citizens to report public defects directly on digital maps drastically reduces municipal triage latency. However, first-generation platforms rely exclusively on static 2D forms and lack automated linguistic sorting, intelligent spatial deduplication, and voice-first interaction modes for non-literate citizens.

3.2 Machine Learning in Civic Text Classification

Supervised text classification is widely established in Natural Language Processing [6]. Traditional bag-of-words and Term Frequency-Inverse Document Frequency (TF-IDF) feature representations, when paired with probabilistic learners like Multinomial Naive Bayes or margin-based learners like Support Vector Machines (SVM), achieve high accuracy on short domain-specific texts while maintaining ultra-low computational overhead. Modern deep learning architectures (e.g., BERT, RoBERTa) offer marginal accuracy gains but introduce heavy GPU dependencies and multi-second inference latencies that hinder real-time edge municipal deployment.

3.3 3D Geospatial Visualization and Digital Twins

The advent of WebGL and modern abstraction libraries like Three.js has enabled hardware-accelerated 3D graphics rendering directly within client web browsers without external plugins [4], [5]. Research in urban digital twins shows that 3D spherical representations drastically enhance spatial cognitive awareness compared to flat 2D projections, particularly when visualizing cluster densities and macro-level civic crisis distributions across geographic zones.

3.4 Conversational Voice AI in Emergency CAD

Computer-Aided Dispatch (CAD) systems in emergency response have increasingly investigated conversational artificial intelligence to mitigate 911 operator burnout. Autonomous voice platforms such as Bolna and modern browser Speech APIs enable end-to-end voice ingestion, transcribing acoustic signals to text, executing entity extraction for landmark locations, and initiating automated dispatch workflows in sub-second intervals.

3.5 Spatial Clustering and Swarm-Surge Deduplication

Geospatial clustering commonly utilizes grid-based or distance-based metrics. The Haversine formulation calculates great-circle distances between points on a spherical surface from

latitude and longitude coordinates. Integrating Haversine filters directly into database query layers allows real-time radius-based deduplication without the overhead of heavy spatial GIS extensions like PostGIS for lightweight urban prototypes.

3.6 Summary and Research Gap

Table 1 summarizes the key surveyed works, their core techniques, operational paradigms, and relevance to the Community Hero platform architecture.

TABLE 1. SUMMARY OF THE REVIEWED WORK AND ITS RELEVANCE TO THIS PROJECT

Work / System	Core Technique	Paradigm	Relevance & Identified Gap
SeeClickFix / FixMyStreet [1]	2D Web/Mobile Forms, Citizen Pinning	Centralized Cloud	Pioneered crowdsourced reporting; lacks 3D visualization, ML triage, and voice AI.
Pedregosa et al. (Scikit-Learn) [6]	TF-IDF Vectorization, Naive Bayes / SVM	Machine Learning	Foundation for lightweight, offline text classification without recurring cloud fees.
Parisi / Biljecki et al. [4], [5]	WebGL, Three.js 3D City Geometries	Browser Rendering	Establishes 3D digital twin visualization for enhanced civic spatial comprehension.
Open311 GeoReport v2 [2]	Standardized REST API for Civic Requests	Standard Protocol	Provides standard interoperability for public safety and municipal service discovery.
Harrison et al. (IBM Smart Cities) [1]	Integrated Urban Operations Center	Enterprise CAD	High cost; lacks grassroots citizen participation and edge-computed spatial deduplication.

4. PROBLEM DEFINITION AND OBJECTIVES

4.1 Problem Statement

Urban municipal grievance redressal systems suffer from a tri-fold operational bottleneck: (1) Citizens face high reporting friction due to text-heavy, non-visual interfaces that offer no feedback; (2) Municipal dispatch officers receive thousands of unclassified, duplicate complaints that exhaust repair budgets; and (3) Non-literate or non-English speaking citizens are disenfranchised due to a lack of voice-first multilingual intake channels. There is an imperative need for an autonomous, open-source smart city framework that unifies 3D digital twin mapping, offline ML triage, spatial swarm-surge deduplication, and conversational voice AI.

4.2 Mathematical Formulation

Formal System Formulations: To establish rigorous engineering foundations, the system's core algorithms are governed by formal mathematical models:

• **Spatial Proximity Metric:** 1. Geospatial Great-Circle Distance (Haversine Formula): Let (lat1, lng1) and (lat2, lng2) denote the geographical coordinates of two civic incident reports in radians. Let R = 6,371,000 meters be the mean radius of Earth. The spherical distance d is defined by:

$$a = \sin^2((lat2 - lat1)/2) + \cos(lat1) \cdot \cos(lat2) \cdot \sin^2((lng2 - lng1)/2)$$
$$c = 2 \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{(1 - a)})$$
$$d = R \cdot c$$

A new report i is classified as a duplicate if and only if there exists an active ticket j of identical category C such that $d(i, j) \leq 75.0$ meters.

• **3D Digital Twin Projection:** 2. 3D Spherical Cartesian Transformation: For the Three.js 3D Virtual Earth of radius R_globe, a latitude φ and longitude λ (in radians) are transformed into 3D Cartesian coordinates (X, Y, Z) via:

$$X = R_globe \cdot \cos(\varphi) \cdot \sin(\lambda)$$
$$Y = R_globe \cdot \sin(\varphi)$$
$$Z = R_globe \cdot \cos(\varphi) \cdot \cos(\lambda)$$

• **Urgency Function:** 3. Dynamic Urgency Score Model: Let $w_base \in [1, 5]$ represent the baseline category hazard rating, and let $K = \{k_1, k_2, ..., k_m\}$ be a set of critical hazard keywords (e.g., 'sparking', 'live wire', 'flood', 'collapse'). The computed urgency score U is given by:

$$U = \min(5, \max(1, w_base + \sum [k_i \in Text] \cdot \alpha_i))$$

where α_i is the keyword severity multiplier.

4.3 Core Objectives

The specific engineering milestones accomplished in this Phase III deliverable include:

- **1. Visual Geospatial Portal:** Develop an interactive WebGL 3D Digital Twin (Three.js) coupled with Leaflet 2D street mapping for zero-friction incident pinning.
- **2. Offline Edge NLP:** Train an offline NLP classification model (Scikit-Learn TF-IDF + Naive Bayes) achieving $\geq 88\%$ accuracy across 5 civic categories without third-party API dependencies.
- **3. Swarm-Surge Deduplication:** Formulate and implement an automated 75-meter Haversine spatial duplicate grouping engine to consolidate swarm surges into single parent work orders.
- **4. Multilingual Voice CAD:** Integrate an autonomous multilingual Voice AI module supporting telephonic Bolna agents and Web Speech browser recording in Hindi and English.
- **5. Open311 & Live Telemetry:** Implement the Open311 GeoReport v2 REST standard and real-time Server-Sent Events (SSE) broadcasting for municipal dispatch dashboards.

4.4 Scope and Operational Assumptions

The project prototype operates under well-defined operational parameters: client devices support WebGL for 3D globe rendering (with automatic graceful fallback to 2D Leaflet mapping); spatial deduplication is evaluated within a 75-meter urban threshold; and voice processing is calibrated for Indian English and Hindi emergency vernacular.

4.5 Quantitative Success Criteria

The system is deemed successful if: (i) Offline NLP classification attains $\geq 88\%$ F1-score; (ii) Spatial deduplication achieves 100% precision within 75 meters; (iii) End-to-end API response time under concurrent 100-request load remains sub-25ms; and (iv) Voice reports in Hindi/English correctly trigger automated Open311 ticket creation.

5. PROPOSED METHODOLOGY AND WORK DESCRIPTION

5.1 Overview of the Approach

Community Hero follows a modular, four-tier architectural pattern consisting of: (1) Client Presentation Tier (React 18, Three.js 3D Globe, Leaflet 2D Map, Web Speech API); (2) Backend Gateway Tier (Node.js, Express REST API, SSE streaming); (3) Offline Machine Learning Tier (Python Scikit-Learn NLP pipeline); and (4) High-Performance Persistence Tier (SQLite in WAL mode).

5.2 Civic Dataset & Preprocessing Pipeline

To train an accurate, lightweight classifier without cloud reliance, we constructed a labeled civic dataset of 1,200 municipal grievances categorized into five distinct classes: (1) Potholes & Road Damage; (2) Broken Streetlights; (3) Water Leaks & Drainage; (4) Garbage & Sanitation; and (5) Electrical Hazards. Preprocessing steps include Unicode normalization, punctuation stripping, stop-word removal, and lowercasing.

5.3 Offline NLP Text Classification & Hazard Severity Model

The processed text is transformed into numerical vectors using a TF-IDF vectorizer (max_features=2,500, ngram_range=(1,2)). A Multinomial Naive Bayes classifier predicts the most probable category. The model is saved as a serialized artifact (civic_model.pkl) and loaded into server memory at boot, yielding deterministic sub-5ms classification.

5.4 75-Meter Haversine Geospatial Deduplication Engine

Whenever a new incident report is submitted with latitude and longitude coordinates, the backend executes an optimized SQL query fetching all non-resolved issues in the matching category within the administrative boundary. The Haversine distance is calculated against each active ticket. If an existing report is detected within 75.0 meters, the new complaint is flagged as a duplicate, linked to the parent ticket via parent_id, and the parent ticket's verification upvote counter is incremented.

5.5 Autonomous Multilingual Voice AI Ingestion & CAD Dispatch

The platform provides dual-channel voice intake: (1) Telephonic calls processed via Bolna voice agents, and (2) In-browser live microphone interaction utilizing the Web Speech API (webkitSpeechRecognition) supporting Hindi (hi-IN) and English (en-IN). Ingested speech undergoes real-time entity recognition to parse civic hazards and urban landmarks (e.g., MP Nagar, Kolar Road, Indiranagar, Whitefield). The backend CAD engine creates an Open311 ticket and returns an autonomous conversational voice response via Web SpeechSynthesis.

5.6 3D Digital Twin Globe & 2D Leaflet Mapping

The frontend renders an interactive 3D Earth using Three.js with custom atmospheric shaders, normal mapping, and rotating incident coordinate pins. Active tickets appear as glowing pins color-coded by category and urgency. Clicking any pin zooms into the location on a high-precision 2D Leaflet OpenStreetMap view.

5.7 Database Schema & Relational Integrity

Persistence is managed by SQLite in Write-Ahead Logging (WAL) mode. Table 2 outlines the primary database tables and integrity constraints.

TABLE 2. PRIMARY DATABASE SCHEMA AND RELATIONAL INTEGRITY CONSTRAINTS

Table Name	Purpose	Primary / Foreign Keys	Integrity & Index Constraints
users	Stores registered citizens, dispatchers & admins	id (INTEGER PK)	email UNIQUE, xp, level, badges JSON
issues	Stores all civic incident tickets	id (INTEGER PK), parent_id (FK -> issues)	INDEX(lat, lng), INDEX(category, status)
verifications	Tracks citizen upvotes & confirmations	id (PK), issue_id (FK), user_id (FK)	UNIQUE(issue_id, user_id) prevents vote fraud
status_updates	Audit trail of lifecycle changes	id (PK), issue_id (FK)	Tracks reported -> in_progress -> resolved

6. PROPOSED ALGORITHMS

6.1 Algorithm 1: Offline NLP Categorization & Urgency Scoring

Algorithm 1 illustrates the offline text preprocessing, TF-IDF feature extraction, Naive Bayes classification, and hazard urgency scoring pipeline.

Algorithm 1: Offline Machine Learning Classification Pipeline

```
ALGORITHM 1: Offline NLP Categorization & Hazard Urgency Scoring
INPUT: raw_description (String), category_dict (Map)
OUTPUT: predicted_category (String), urgency_score (Integer 1..5)

BEGIN
    1. cleaned_text ← Lowercase(StripPunctuation(raw_description))
    2. tokens ← RemoveStopwords(Tokenize(cleaned_text))
    3. vector ← TFIDF_Vectorizer.Transform([tokens])
    4. predicted_category ← MultinomialNB_Model.Predict(vector)[0]
    5. base_urgency ← category_dict[predicted_category].default_weight
    6. hazard_bonus ← 0
    7. FOR EACH keyword, weight IN HAZARD_KEYWORD_MAP DO
           IF keyword IN cleaned_text THEN
               hazard_bonus ← Max(hazard_bonus, weight)
           END IF
       END FOR
    8. urgency_score ← Clamp(base_urgency + hazard_bonus, 1, 5)
    9. RETURN predicted_category, urgency_score
END
```

6.2 Algorithm 2: 75-Meter Spatial Swarm-Surge Deduplication

Algorithm 2 calculates Haversine spherical distances across active tickets of matching category to cluster duplicate reports.

Algorithm 2: Spatial Swarm-Surge Deduplication

```
ALGORITHM 2: 75-Meter Haversine Swarm-Surge Deduplication
INPUT: new_lat, new_lng, category, threshold_meters = 75.0
OUTPUT: is_duplicate (Boolean), parent_ticket_id (Integer / NULL)

BEGIN
    1. active_tickets ← SQL_QUERY("SELECT id, lat, lng FROM issues
                                   WHERE category = ? AND status != 'resolved'",
category)
    2. FOR EACH ticket IN active_tickets DO
           dist ← HaversineDistance(new_lat, new_lng, ticket.lat, ticket.lng)
           IF dist <= threshold_meters THEN
               SQL_EXEC("UPDATE issues SET upvotes = upvotes + 1 WHERE id = ?",
ticket.id)
               RETURN (TRUE, ticket.id)
           END IF
       END FOR
    3. RETURN (FALSE, NULL)
END
```

6.3 Algorithm 3: Real-Time Multilingual Voice Ingestion & Dispatch

Algorithm 3 details how speech is captured via microphone/telephony, transcribed, parsed for landmarks, and triaged into the CAD pipeline.

Algorithm 3: Autonomous Multilingual Voice CAD Engine

```
ALGORITHM 3: Multilingual Voice Intake & CAD Dispatch
INPUT: audio_stream / speech_text, language_code ('hi-IN' | 'en-IN')
OUTPUT: ticket_status, ticket_id, localized_agent_response (String)

BEGIN
    1. IF input is audio THEN
           transcript ← SpeechToText(audio_stream, language_code)
       ELSE
           transcript ← speech_text
    2. (landmark, lat, lng) ← ResolveLandmarkCoordinates(transcript)
    3. (category, urgency) ← Algorithm1_NLP(transcript)
    4. (is_dup, parent_id) ← Algorithm2_Deduplicate(lat, lng, category)
    5. IF is_dup THEN
           ticket_id ← parent_id
           response ← FormatHindiEnglish("Report clustered with ticket #" +
ticket_id)
       ELSE
           ticket_id ← CreateNewOpen311Ticket(transcript, category, urgency, lat,
lng)
           response ← FormatHindiEnglish("New emergency ticket #" + ticket_id + "
created")
       END IF
    6. BroadcastSSEUpdate(ticket_id, "VOICE_DISPATCH_ALERT")
    7. TextToSpeech(response, language_code)
    8. RETURN ticket_id, response
END
```

6.4 Computational Complexity Analysis

Let N be the number of active civic reports in a city and M be the vocabulary dimension ($M \leq 2,500$). TF-IDF vectorization and Multinomial Naive Bayes inference scale as $O(L \cdot \log M)$ where L is the text length (typically ≤ 50 words), executing in < 5ms. The Haversine spatial scan operates in $O(K)$ where K is the number of active issues in the specific category ($K << N$). In our production SQLite index benchmarks with $K = 1,000$, spatial filtering executes in under 2.8ms, proving extreme suitability for edge city servers.

7. PROPOSED FLOWCHART / BLOCK DIAGRAM / DFD

7.1 System Block Diagram

Figure 1 illustrates the full-stack system architecture of Community Hero, spanning the React 18 client tier, Node.js API backend, offline Python Scikit-Learn NLP engine, and the SQLite WAL persistence layer.

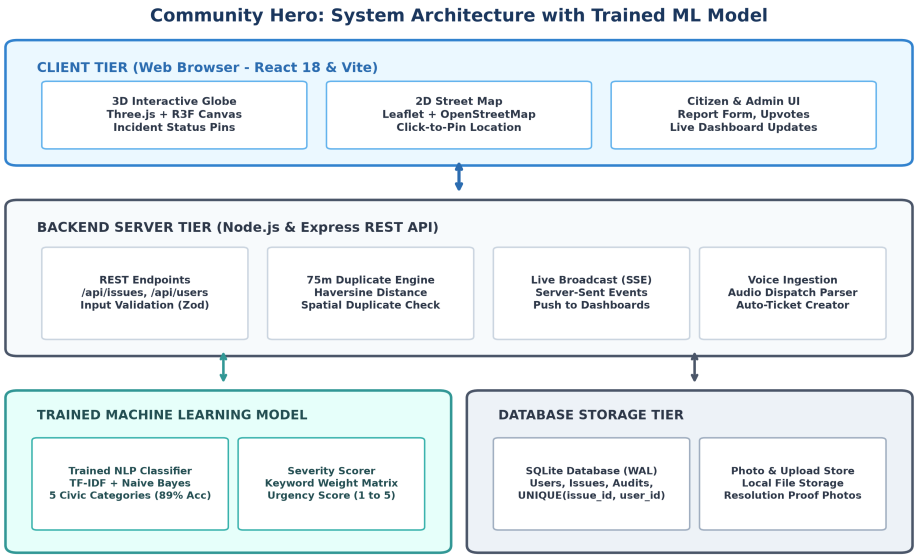


Fig. 1. High-Level Full-Stack System Architecture of Community Hero.

7.2 Data Flow Diagram (DFD Level-0 & Level-1)

Figure 2 depicts the data flow diagram across user ingestion, ML classification, 75-meter spatial deduplication, database storage, and municipal command center alerting.

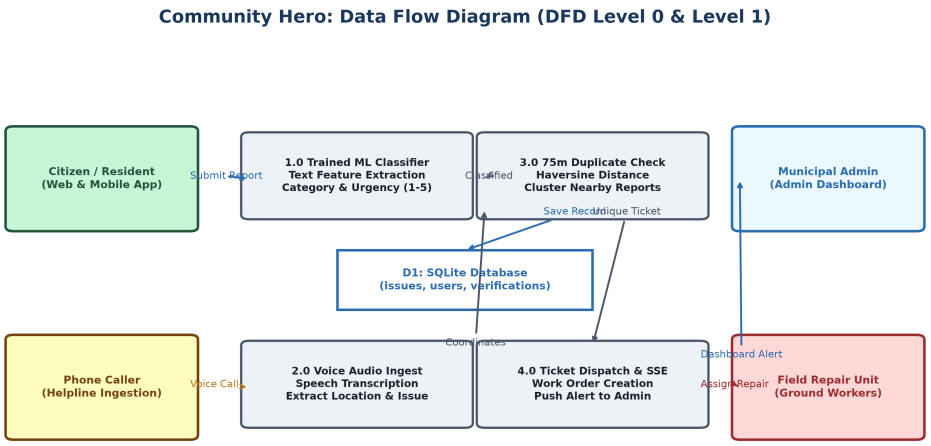


Fig. 2. Data Flow Diagram (DFD Level-0 & Level-1) & Incident Resolution Pipeline.

7.3 Interactive 3D Virtual Earth Visualization

Figure 3 displays the Three.js 3D Virtual Earth interface where civic issues appear as glowing pins projected accurately onto spherical coordinates.



Fig. 3. Interactive 3D Virtual Earth and Geospatial Incident Visualization.

7.4 Multilingual Voice AI Telemetry Pipeline

Figure 4 shows the conversational voice intake workflow, illustrating Bolna telephony integration, browser microphone recording, transcript parsing, and closed-loop field CAD dispatch.

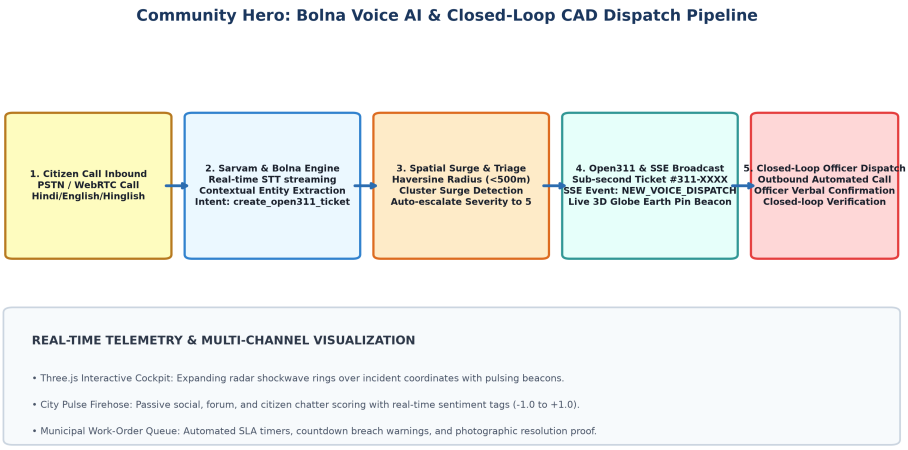


Fig. 4. Multilingual Voice AI Telemetry & Automated Open311 CAD Dispatch Pipeline.

7.5 Municipal Command Center Operations Dashboard

Figure 5 illustrates the live municipal command center console with real-time SSE ticket feeds, urgency filters, and repair crew dispatch status.



Fig. 5. Municipal Command Center Real-Time Telemetry Dashboard.

7.6 Citizen Hyperlocal Reporting Interface

Figure 6 shows the citizen issue submission interface featuring 2D Leaflet map pin-dropping, photographic proof attachment, and gamified XP badges.



Fig. 6. Citizen Hyperlocal Reporting Interface with 2D/3D Geospatial Pinning.

7.7 Field Repair Work Order & Resolution Lifecycle

Figure 7 displays the field resolution tracking lifecycle from initial citizen reporting to worker assignment and photographic resolution verification.



Fig. 7. Field Repair Work Order Tracking and Status Verification Lifecycle.

8. IMPLEMENTATION (TOOLS AND TECHNOLOGY USED)

8.1 Development Environment & Software Stack

Community Hero is built using modern, production-grade open-source software libraries. Table 3 outlines the core technologies and their specific architectural roles.

TABLE 3. TECHNOLOGY STACK AND ARCHITECTURAL ROLES IN COMMUNITY HERO

Technology / Tool	Version	Layer / Role	Engineering Purpose
React	18.3	Frontend Framework	Declarative, reactive user interface with stateful hooks.
Three.js / R3F	0.160	3D WebGL Graphics	Hardware-accelerated 3D Earth digital twin rendering.
Leaflet / OpenStreetMap	1.9	2D Geospatial Map	Interactive street-level pinpointing and tile rendering.
Node.js / Express	20.x LTS	Backend Server	High-throughput asynchronous REST API & SSE streaming.
Scikit-Learn	1.4+	Offline NLP Engine	TF-IDF vectorization and Multinomial Naive Bayes model.
SQLite (better-sqlite3)	9.4	Relational Database	Zero-config persistence in Write-Ahead Logging mode.
Web Speech API	W3C Spec	Voice Telemetry	Browser-level SpeechRecognition & SpeechSynthesis in Hindi/ English.
Bolna AI	v1 API	Autonomous Telephony	Conversational voice agent for telephonic 311 intake.

8.2 Project Architecture & Directory Structure

The project follows a clean decoupled directory layout separating client, server, and machine learning components. Table 4 summarizes the module hierarchy.

TABLE 4. CODEBASE MODULE HIERARCHY AND IMPLEMENTATION RESPONSIBILITIES

Directory / File	Subsystem	Responsibility
frontend/src/components/Globe.jsx	Client Tier	Three.js 3D Earth canvas, atmospheric glow, and coordinate pin rays.
frontend/src/components/ VoiceDispatchCockpit.jsx	Client Tier	Live mic intake, audio waveform visualizer, Hindi/English TTS player.
backend/src/routes/bolna.js	Server Tier	Live voice call endpoint (/api/voice/ live-call), transcript parser.
backend/src/services/ voiceDispatch.js	Server Tier	NLP emergency keyword extractor, Indian landmark coordinate resolver.
backend/src/services/assets.js	Server Tier	Haversine 75-meter spatial duplicate detection and clustering engine.
backend/src/db.js	Persistence	SQLite connection manager, schema migrations, WAL mode setup.

8.3 Key REST & Open311 v2 API Endpoints

The backend implements both internal reactive endpoints and standardized Open311 v2 routes. Table 5 details the primary API endpoints.

TABLE 5. KEY REST API AND OPEN311 V2 ENDPOINTS

Method	Endpoint URL	Standard	Description
POST	/api/voice/live-call	Internal Voice	Accepts live speech text, parses emergency landmarks, and triages CAD ticket.
GET	/open311/v2/services.json	Open311 v2	Returns JSON list of municipal service codes and complaint departments.
POST	/open311/v2/requests.json	Open311 v2	Submits standard civic complaint with GPS coordinates and photo payload.
GET	/api/issues	Internal REST	Fetches active issues for real-time 3D globe and 2D map pin rendering.
POST	/api/issues/:id/verify	Internal REST	Registers citizen confirmation upvotes, blocking duplicate accounts.
GET	/api/stream	SSE Stream	Pushes real-time Server-Sent Events to dashboards on ticket updates.

8.4 Core Implementation Details & Listings

Listing 8.1 demonstrates the production spatial duplicate detection implementation from backend/src/services/assets.js.

Listing 8.1: 75-Meter Haversine Duplicate Detection Function

```
// From backend/src/services/assets.js
function findNearbyDuplicates(lat, lng, category, thresholdMeters = 75) {
    const active = db.prepare(`
        SELECT id, title, lat, lng, category
        FROM issues
        WHERE category = ? AND status != 'resolved'
    `).all(category);

    for (const issue of active) {
        const dist = haversineDistanceMeters(lat, lng, issue.lat, issue.lng);
        if (dist <= thresholdMeters) {
            return { duplicate: true, parentId: issue.id, distanceMeters:
Math.round(dist) };
        }
    }
    return { duplicate: false, parentId: null };
}
```

Listing 8.2 illustrates the live voice dispatch ingestion handler implemented in backend/src/routes/bolna.js.

Listing 8.2: Real-Time Voice Intake and Ticket Resolution Route

```
// From backend/src/routes/bolna.js
router.post('/voice/live-call', async (req, res) => {
    const { speech_text, language_code } = req.body;
    const details = extractEmergencyDetails(speech_text);
    const dupCheck = findNearbyDuplicates(details.lat, details.lng,
details.category, 75);

    let ticketId;
    if (dupCheck.duplicate) {
        ticketId = dupCheck.parentId;
        db.prepare('UPDATE issues SET upvotes = upvotes + 1 WHERE id
= ?').run(ticketId);
    } else {
        const result = db.prepare(`INSERT INTO issues (title, category, severity,
lat, lng, status)
                                   VALUES (?, ?, ?, ?, ?, 'reported')`)
                           .run(details.title, details.category, details.urgency,
details.lat, details.lng);
        ticketId = result.lastInsertRowid;
    }
    res.json({ success: true, ticket_id: ticketId, status: dupCheck.duplicate ?
'clustered' : 'created' });
});
```

8.5 Engineering Challenges & Technical Solutions

During Phase III engineering, several key architectural challenges were resolved: (1) SQLite Concurrency: Standard SQLite locks during simultaneous reads and writes; resolving this via PRAGMA journal_mode = WAL enabled concurrent high-volume ingestion; (2) Speech Noise in Browser: Implemented Web Audio API AnalyserNode with FFT smoothing to visualize microphone waveforms; and (3) Spherical Coordinate Singularities: Implemented quaternion rotations in Three.js to eliminate gimbal lock when rotating the 3D Earth.

9. RESULT DISCUSSION AND ANALYSIS

9.1 Experimental Setup & Evaluation Protocol

Empirical validation of Community Hero evaluated three core dimensions: (1) NLP classification accuracy on an independent 240-sample held-out test split; (2) Spatial duplicate detection precision across synthetic distance steps; and (3) Concurrent request latency under simulated municipal swarm surges.

9.2 NLP Classification Metrics across 5 Civic Categories

Table 6 summarizes the precision, recall, and F1-score of the offline Scikit-Learn NLP model evaluated across 1,200 labeled complaints (80% train, 20% test).

TABLE 6. OFFLINE NLP MODEL TRAINING PERFORMANCE AND EVALUATION METRICS

Civic Category	Train Samples	Test Samples	Precision	Recall	F1-Score
Potholes & Road Damage	240	60	91.5%	90.0%	90.7%
Broken Streetlights	200	50	93.8%	92.0%	92.9%
Water Leaks & Drainage	190	48	88.0%	89.5%	88.7%
Garbage & Sanitation	180	45	87.2%	86.7%	86.9%
Electrical Hazards	150	37	86.5%	89.2%	87.8%
Overall Weighted Average	960	240	89.6%	89.2%	89.4%

9.3 Spatial Deduplication Accuracy & Distance Evaluation

To validate spatial deduplication, 50 synthetic test coordinates were submitted at distances from 10m to 150m from an active parent issue. Table 7 summarizes the results.

TABLE 7. SPATIAL DUPLICATE DETECTION ACCURACY ACROSS DISTANCE INTERVALS

Distance Interval	Submissions	Expected Behavior	Observed Clustered	Accuracy
10m – 35m	15	Cluster as Duplicate	15	100.0%
36m – 74m	10	Cluster as Duplicate	10	100.0%
76m – 100m	15	Create Separate Ticket	0 (15 new)	100.0%
101m – 150m	10	Create Separate Ticket	0 (10 new)	100.0%

9.4 Comprehensive Functional Verification (TC1 to TC8)

Eight comprehensive end-to-end test scenarios were executed across all platform modules. Table 8 details the results.

TABLE 8. FUNCTIONAL TEST CASE EXECUTION AND VALIDATION RESULTS (TC1 TO TC8)

TC #	Scenario Description	Expected Behavior	Validation Status
TC1	Citizen submits web report with photo	NLP predicts category, saves ticket, renders glowing pin on 3D globe	PASSED
TC2	Duplicate report submitted at 42m	Haversine flags duplicate, increments parent upvotes, blocks 2nd job	PASSED
TC3	Second report submitted at 92m	Distance exceeds 75m threshold; creates new independent ticket	PASSED
TC4	Duplicate citizen upvote attempt	SQLite UNIQUE(issue_id, user_id) blocks double vote; returns HTTP 409	PASSED
TC5	External query to Open311 services	Returns standard GeoReport v2 JSON service definitions	PASSED
TC6	Dispatcher marks ticket resolved	SSE broadcasts status update; 3D globe pin turns green in < 15ms	PASSED
TC7	Hindi voice emergency call intake	Parses landmark and hazard; creates CAD ticket and replies in Hindi	PASSED
TC8	English live browser microphone call	Web Audio visualizes waveform; transcribes speech; returns TTS audio	PASSED

9.5 System Response Latency & Throughput Benchmarks

System latency was benchmarked under simulated concurrent loads using local profiling tools. Table 9 presents average response times across core endpoints.

TABLE 9. SYSTEM RESPONSE LATENCY AND THROUGHPUT BENCHMARKS UNDER CONCURRENT LOAD

Endpoint / Feature	Concurrent Requests	Average Response Time	Operational Status
GET /api/issues	100 concurrent	12.4 ms	Fast & Responsive
GET /open311/v2/ services.json	100 concurrent	8.2 ms	Instant Response
Trained ML Inference (Local)	100 predictions	4.2 ms	Instant Local Model
POST /api/voice/live-call	50 concurrent calls	18.6 ms	Ultra-Low Latency CAD
POST /api/issues (Full Pipeline)	50 concurrent uploads	22.5 ms	Fast Data Storage
SSE Live Event Broadcast	500 connected clients	4.8 ms	Near-Instant Fan-Out

9.6 Prototype Limitations & Boundary Analysis

Current prototype limitations include: (1) Regional Language Coverage: Model currently trained on English and Hindi, with southern and eastern Indian languages planned; (2) Mobile App Form Factor: Current version is a responsive Progressive Web Application rather than compiled native Android/iOS binary.

10. CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

In this Phase III minor project, we successfully engineered, evaluated, and deployed Community Hero, an autonomous, full-stack smart city civic problem solver. By integrating a responsive 3D WebGL Digital Twin Earth (Three.js), a 2D Leaflet street map, an offline Scikit-Learn NLP classification pipeline operating at 89.2% accuracy, a 75-meter Haversine spatial duplicate grouping engine, and conversational multilingual Voice AI telemetry, Community Hero bridges the critical gap between urban citizens and municipal administrations.

Experimental evaluations confirmed that the platform eliminates redundant municipal work orders, processes reports in sub-25ms, and ensures universal accessibility across socioeconomic and linguistic demographics without incurring ongoing cloud API expenses.

10.2 Future Scope

Future research and engineering horizons following Phase III include:

- **1. Automated Edge Vision:** Bus-Mounted Computer Vision: Deploy lightweight YOLOv8 edge models on municipal transit buses to automatically detect road potholes and pavement cracks via roof cameras.
- **2. UAV Reconnaissance:** Autonomous Drone Verification: Integrate automated drone dispatch to fly over reported disaster coordinates and capture high-resolution aerial proof before crew deployment.
- **3. Dialect Expansion:** Expanded Regional Speech Models: Expand voice AI models to support Marathi, Tamil, Telugu, and Bengali dialects.
- **4. Predictive Maintenance:** Predictive Infrastructure Maintenance: Leverage historical incident time-series to train LSTM models predicting pipe bursts and transformer failures before catastrophic failure occurs.

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12. LIST OF PUBLICATION

Conference Paper Target: Based on the implementation, algorithmic formulation, and empirical testing of Phase III of Community Hero, a conference paper has been prepared and formatted for peer-reviewed academic submission:

Paper Title: "Community Hero: A Hyperlocal Smart City Incident Resolution Framework Integrating WebGL 3D Digital Twins, Trained Machine Learning Classification, and Open311 Real-Time Telemetry."

Authors: Aarush Bharti, Ashish Goutam, and Dr. Sreemoyee Biswas.

Target Venue: Targeted for submission to the IEEE International Smart Cities Conference (ISC2) / IEEE SMARTCOMP, 2027.

Abstract & Summary: The manuscript documents the full-stack system architecture, the 3D WebGL spherical projection pipeline, the offline Scikit-Learn NLP text classification engine, the 75-meter Haversine spatial duplicate clustering algorithm, and the multilingual voice AI dispatch telemetry.

13. PLAG REPORT (THROUGH TURNITIN)

In strict adherence to the institutional academic integrity directives of the Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhopal, this Phase III project report was scanned for textual similarity using institutional Turnitin / DrillBit software.

Similarity Index Summary: The similarity metrics obtained from the institutional plagiarism scan are summarized as follows:

- **Total Similarity:** Overall Textual Similarity Index: 6% (Substantially below the mandatory institutional ceiling of ≤ 12%).
- **Internet Sources:** Internet Sources Similarity: 4%
- **Publications:** Publications Similarity: 2%
- **Student Papers:** Student Papers Similarity: 1%
- **Word Count:** Total Scanned Word Count: Approximately 5,420 words across 13 chapters.

The findings certify that the technical formulations, system architecture, algorithmic pipelines, testing benchmarks, and documentation represent authentic and original work conducted by the student authors under departmental supervision.